

Chapter 4

Convex Structure

4.1 Convex optimization and Lagrange duality

Admittedly, a section about optimization techniques is a bit off topic. However, convex optimization and, in particular, *semidefinite programming*¹ turned out to be an extremely useful tool in the context of quantum information theory, so we will at least sketch the basic framework here.

The utility of these methods is threefold: (i) they provide powerful algorithms to solve problems numerically. In contrast to many other numerical techniques these algorithms are certifiable. That is, they return a result together with a (usually negligible) error bound which is essentially enabled by duality theory. (ii) Duality theory allows to convert numerical results into analytic proofs and often enough gives some analytic insight (if not a solution) without exploiting any silicon brain. (iii) Having expressed a problem in terms of e.g. a semidefinite program may be considered as a solution of the problem in very much the same way as we consider a reduction to an eigenvalue problem to be a solution. Moreover, it enables a classification of the problems complexity in the sense of complexity theory since interior point methods provide a general polynomial-time solution.

Conic programs Consider finite-dimensional real Hilbert spaces $\mathcal{V}$ and $\mathcal{V}'$. These may be bare $\mathbb{R}^d$'s but could also be spaces of Hermitian matrices or spaces of polynomials of a certain degree. Suppose $\mathcal{K} \subset \mathcal{V}$ is a closed convex cone which is pointed ($\mathcal{K} \cap -\mathcal{K} = 0$) and has a non-empty interior. For any linear map $T : \mathcal{V} \rightarrow \mathcal{V}'$ and $c \in \mathcal{V}$, $b \in \mathcal{V}'$ the optimization problem

$$\inf_{x \in \mathcal{K}} \{ \langle c|x \rangle \mid T(x) = b \} =: C_p \quad (4.1)$$

¹Depending on what community one comes from it needs some time to get used to the terminology: conic, semidefinite or linear *programs* are just optimization problems of a certain type.

is called a *convex conic program*. Assigned to this is a *dual problem*² which reads

$$\sup_{y \in \mathcal{V}'} \{ \langle b|y \rangle \mid c - T^*(y) \in \mathcal{K}^* \} =: C_d, \quad (4.2)$$

where T^* is the dual map (i.e., $\langle T^*(y)|x \rangle = \langle y|T(x) \rangle$) and $\mathcal{K}^*$ is the dual cone defined by $\mathcal{K}^* := \{x \in \mathcal{V} \mid \forall z \in \mathcal{K} : \langle z|x \rangle \geq 0\}$.

It holds in general that $C_p \geq C_d$ – this is called *weak duality*. *Strong duality*, meaning $C_p = C_d$, holds if $T(x) = b$ is fulfilled for some x which is in the interior of $\mathcal{K}$. In this case the problem in Eq.(4.1) is called *strictly feasible* and the supremum C_d is attained in the dual. Similarly, strong duality holds if the dual problem in Eq.(4.2) is strictly feasible (i.e., $c - T^*(y)$ is in the interior of $\mathcal{K}^*$ for some y). In this case the infimum C_p is attained in Eq.(4.1).

For many conic programs there exist polynomial-time algorithms. Roughly speaking, the requirements for that are that membership in the cone can be decided efficiently (like for the cone of positive semidefinite matrices) and that bounds on the norm of candidate solutions can be given.

Semidefinite programs are special instances of convex conic programs. Here T is a linear map between spaces of Hermitian matrices and $\mathcal{K} = \mathcal{K}^*$ is the cone of positive semidefinite matrices. Traditionally, semidefinite programs are phrased in a slightly different way than in Eqs.(4.1,4.2). The weak duality inequality is usually stated as

$$\inf_{X \geq 0} \{ \text{tr}[F_0 X] \mid \text{tr}[F_i X] = b_i \} \geq \sup_{y \in \mathbb{R}^n} \left\{ \langle b|y \rangle \mid F_0 \geq \sum_{i=1}^n y_i F_i \right\}, \quad (4.3)$$

where $b \in \mathbb{R}^n$, $F_0, F_1, \dots, F_n$ as well as X are Hermitian matrices and “ $\geq$ ” for matrices denotes the standard matrix ordering induced by the cone of positive semidefinite matrices. Strong duality, meaning equality in Eq.(4.3), holds if at least one of the two optimization problems is strictly feasible. That is, if there is either an $X > 0$ with $\text{tr}[F_i X] = b_i$ or a y such that $F_0 > \sum_i y_i F_i$. As mentioned in the previous paragraph, if one of the problems is strictly feasible, then the extremum in the dual problem is attained.

If the extrema in Eq.(4.3) are both attained and equal (i.e., in particular strong duality holds) then a simple condition called *complementary slackness* relates optimal solutions X' and y' :

$$[F_0 - \sum_i y'_i F_i] X' = 0. \quad (4.4)$$

²The relation between the *primal problem* in (4.1) and the dual problem (4.2) is more symmetric than it appears at first sight. In particular, the dual problem is a convex conic program in its own right. To see the symmetric structure note that $T(x) = b$ defines an affine set which can be parametrized by $\tilde{y} \in \mathbb{R}^n$ via $x = e_0 + \sum_{i=1}^n \tilde{y}_i e_i$ with suitable $e_i \in \mathcal{V}$. Defining further $\tilde{b}_i := \langle c|e_i \rangle$ we can rewrite the primal problem without equality constraints in a form similar to that of Eq.(4.2) up to a constant offset (since $\langle \tilde{b}|\tilde{y} \rangle = \langle c|x \rangle - \langle c|e_0 \rangle$) and a sign which has to be spent to turn the inf into a sup.

That is, X' and $[F_0 - \sum_i y'_i F_i]$ have to have orthogonal supports. Moreover, any y' is then optimal iff there exists an $X' \geq 0$ such that Eq.(4.4) holds and the constraints $\text{tr}[F_i X'] = b_i$ and $F_0 \geq \sum_i y'_i F_i$ are satisfied.

4.2 Literature

Convex optimization is described in detail in the textbook [?]. [?] is a good source for more on semidefinite programs and more on conic programs, in particular about complexity related questions, can be found in [?].

